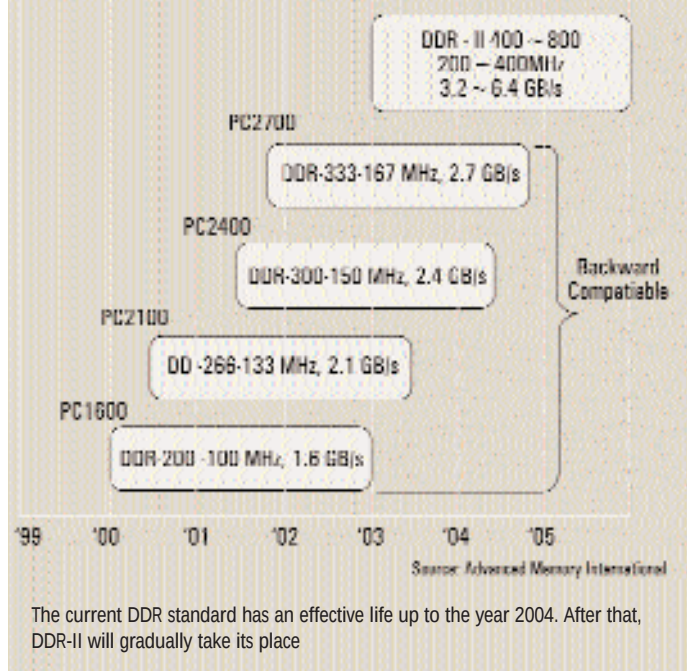


DDR Performance Migration Path



to another has been fairly graceful, with all companies getting behind one standard. But in the current scenario, manufacturers have had to figure out what technology is likely to be accepted, or they might well find themselves saddled with a product that won't sell or will become defunct.

The year 2001 too had probably seen the most confusing times when it came to PC technology. For any company or individual considering buying a new PC, the thought that it might get obsolete or defunct in just a couple of months is frightening. This amazing dilemma of choosing the right memory interface will continue to cause more heartburn because the new year will bring in even greater changes in computer hardware, so much so that we will see processors advancing over 3,000 MHz in raw speed! These processors, however, will have a tough time communicating since memory speeds and interfaces have failed to make parallel improvements because of incessant squabbling over memory standards. No one standard has been universally approved or recommended.

At present, DDR-SDRAM and RDRAM are the major competitors vying for the title of 'Memory of the Future'. RDRAM was seen by most as the future of memory. Intel promoted it, but things did not go as per their plan. With a buggy chipset and low chip yields, it failed to take off and never became mainstream.

DDR-SDRAM, on the other hand, is currently going strong and is estimated to jump to a

massive 50 per cent of all the memory to be shipped in the year 2002 and is all set to become the dominant memory technology of the future. This is a remarkable jump despite the fact that the Intel Pentium 4 is currently incapable of running alongside DDR-SDRAM because of unavailable chipsets. Also in its favour is the price—it is comparable to that of the run-of-the-mill SDRAM. Current prices show a difference of less than Rs 1,000 on the same size modules, which is impressive for a newer memory technology.

The DDR-SDRAM approach is a simpler, more evolutionary approach, while Rambus proposes to revolutionise PC design. The advantages of the evolutionary approach of DDR-SDRAM are simple and easy to under-

stand. It allows for a smooth transition from current technologies and is an open standard, which most importantly means that any and every manufacturer can manufacture and use it. But revolutionary changes involved in the transition to RDRAM would mean taking a huge risk, because the effects will be across the entire system architecture.

DDR technology has already breathed new life into the AMD processor and will do so with the Pentium 4 too. Whether or not it will replace all the solutions in the market by becoming the only dominant memory solution is yet to be seen. Signs are that it will.

Intel will soon release its mainstream DDR solution, code-named Brookdale-D, for the Pentium 4. Further on, we have the infamous VIA P4X266 chipset for the Pentium 4 supporting DDR.

Also riding strong is the AthlonXP along with new platform solutions from nVidia for the PC market supporting DDR. We already have DDR 333 being sampled and motherboards for it being readied.

All in all, it would be safe to say that DDR has matured enough from combating with RDRAM to become the only viable solution for the Pentium 4 and AthlonXP. Nevertheless, for DDR-SDRAM to succeed in numbers it will need the Pentium 4. The extreme loss of bandwidth that the 845-based solution is offering the Pentium 4 with its ageing SDRAM solution will soon be replaced and that will probably be the ultimate victory for DDR-SDRAM. ■

Next in Line

Changes and innovations in technology move at a pace usually difficult to keep up with. The Joint Electronic Device Engineering Council (JEDEC) has already started development work on the DDR II. There are more than 120 companies on the committee working on the standard including DRAM industry leaders Elpida, Hynix, IBM Microelectronics, Infineon, Micron, Mitsubishi, Mosel Vitelic, Samsung, Toshiba, as well as chipset and infrastructure leaders AMD, ALi, CST, ServerWorks, VIA, Texas Instruments, and others. DDR II promises significant improvements with another doubling in bandwidth, optimised bus utilisation, and improved power management. All this can be relegated as marketing hype when compared to the specifications that JEDEC is proposing—DDR 400 (PC 3200, 3.2 GBps) and DDR 533 (PC 4300, 4.3 GBps). This is simply mind-numbing and when paired in dual channel combinations, the result will be outrageous.